

VRDots Pilot — Write-Up Index

All pilot sessions · n = 1 observer · Meta Quest 3 · 2026-03-23 – 2026-04-02

File · Title	Sessions	Pages	Key Findings
baseline_writeup.pdf Baseline — No Depth, No Swap	260323_1534 (pre-v0.2.0) · 260324_0716 (v0.2.0)	22	+50 pp cueing (S1); +28 pp (S2, cursor-jump artifact). Red/green accuracy asymmetry in S1 likely reflects imperfect isoluminance. Both sessions confirm reliable delayed-onset cueing in VR.
swap_writeup.pdf Flat Swap Conditions — Motion Swap & Dots50 Swap	260324_1010 (MotionSwap) · 260325_1039 (Dots50Swap)	20	Motion Swap (100%) attenuates cueing from ~27 pp → ~16 pp. Dots50 Swap (50% noise-subfield reassignment) produces no attenuation (~34 pp). Temporal onset asymmetry alone provides residual cueing benefit.
depth_baseline_writeup.pdf Depth Baseline — No Swap, 0.03 m & 0.10 m	260325_1831, 260325_1914 (0.10 m) · 260325_2013 (0.03 m)	25	At 0.10 m: Far translation +19–28 pp*; Near translation near chance (2D signal masked by large disparity). At 0.03 m: Far +24 pp*, Near +9 pp n.s. Near-plane floor motivates DepthParam sweep.
depth_swapctrl_writeup.pdf Depth Swap Control — ZdA / ZdB / N, 0.05 m	260330_1853, 260331_0621, 260401_1313, 260401_1349, 260401_1511, 260401_1705 (ZdA) · 260330_2012, 260331_1530 (mono-R) · 260331_1705, 260331_1706, 260331_1707 (ZdB) · 260331_1705, 260331_1706, 260331_1707 (N)	24	ZdA (no floor, no changes in depth plane): +12 pp binocular, +16 pp* binocular, +12 pp* binocular, +16 pp* binocular. ZdB (noise dot changes plane): +16 pp** binocular, +12 pp* binocular. N (no swap): +10 pp* binocular. Far > Near (binocular N); Near/Far difference entirely stereoscopic (disappears monocularly).
depthparam_writeup.pdf Depth Parameter Sweep — 0.03 / 0.05 / 0.10 / 0.15 m	260402_0624 (0.10 m) · 260402_0656 (0.15 m) · 260402_0716 (0.03 m) · 260402_0757 (0.05 m)	21	Far translation cueing: +47 pp across all depths (consistent). Near translation cueing: negative or near zero at all depths — Near-plane performance floor persists even at 0.03 m. Depth separation does not recover Near cueing; floor is likely a stereo-motion masking effect.